



Panel in discussion during webinar

startup and entrepreneurial perspective, deep-tech angle, formal education perspective, and skilling and upskilling angles.

Dr B.K. Murthy, senior director and group coordinator, R&D in IT and Digital India Corporation, MeitY, was the chief guest and keynote speaker at the online event. Niju Vijayam, executive director, Avanteum Advisors LLP, presented the report and was a speaker at the event.

Questions answered

1. What are the challenges in AI and ML electronics hardware vertical?
2. What are the opportunities in AI and ML electronics hardware vertical?
3. What role can corporates, startups, and government play in tapping these opportunities?
4. How big are AI and ML in electronics hardware?
5. What is the market size in India, and what are the key drivers?
6. Does AI and ML electronics hardware have any opportunities for entrepreneurs?
7. How can academia prepare students for the opportunities?
8. Who all can be the catalyst for increased AI hardware implementation?
9. How are the investments in the technology and skills sectors shaping up?
10. What's the road ahead?

"In the early 1980s, there were a lot of promises from IT and ITES verticals.

These promises have been delivered. Almost similar scales of promises are being promised by AI, ML, and similar technologies now. Just like IT and ITES verticals in India have made their presence felt globally, we expect that AI and other emerging technologies will do the same. 'Now' is the time to start focusing more on AI. Even hardware engineers should start learning about AI. Remember that the clients now expect a complete ecosystem of hardware, software, and emerging technologies," said Dr Murthy at the event.

Points to remember

1. The development of the Aarogya Setu App is proof of how AI, ML and data science can benefit everyone through an aggregator platform.
2. AI and data science are the key pillars of the present that are building the future.
3. We can come out of Covid-19 imposed recession using these technologies in a year or two, provided we work together.
4. The total AI hardware workforce will be around 1.16 million by this year end.
5. The amount of computing done is increasing two to three times on average. This is the fastest ever growth.
6. India is still a nascent market in terms of AI, emerging technologies,

and AI/ML electronics hardware development.

7. Just like dot com, AI is an all-encompassing technology. It will take time to mature. We need to be mature about its adoption and funds being pumped into it.

8. Budding engineers need to focus on a specific approach to AI and ML and electronics hardware side of them.

9. AI will be the most dominant technology in the future. More than twenty billion dollars are being spent annually on AI.

"In the next five years, the demand for AI electronics hardware professionals is expected to cross 0.45 million. The rate of growth will be 92 per cent during this period. Demand is expected to be across the spectrum, including manufacturing, agriculture, food tech, transportation to healthcare. It has the potential to contribute to the country's goal of being a five trillion dollars economy," said Poornima Shenoy.

Challenges

1. In India, there is a very low-value addition happening in terms of system integration.
2. We should not be making the same mistake we made for the Internet of Things (IoT). The ecosystem should be created for the world and we should not just focus on consuming what the world creates.
3. The adoption of AI is gated by the